

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1591

James E. Rogers Energy Complex 138 kV
July 03, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1591. The Interconnection Customer proposes a 339.9 MW solar photovoltaic facility interconnecting at the James E. Rogers Energy Complex 138 kV substation in the Pennsylvania area. The request was received on October 09, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 456 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$2,966,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Fair Oaks 345 kV/138 kV TX	Norway Point Hydropower Project to James E. Rogers Energy Complex 138 KV	106.1	625	45.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fair Oaks 345 kV/138 kV TX	Replace transformer with higher-rated unit	\$11,571,000	26% (\$2,966,000)
	Total allocated to project		\$2,966,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.